

# Pure-Weyl gravity programme: executive summary for physicists

**Last substantive update: 22 July 2026 — Phase 1 closed with an exact classification ending: a causal counterflow parent exists, but its declared same-field stationary family supplies no robust physically healthy Phase-2 candidate.**

**Research context.** Asger Alstrup Palm directs the programme and is the accountable human contact. AI systems perform substantial derivation, verification, literature mapping, and drafting. Results are offered as scoped claims with reproducible artifacts. Palm’s DTU affiliation identifies his academic role and does not imply institutional endorsement.

## The sixty-second version

Pure Weyl gravity is a four-dimensional, locally conformal, fourth-order theory of the metric. Relative to Einstein gravity it has additional local gravitational solutions, and conventional pole or oscillator analyses attach indefinite signs to them. The programme starts from the position that a propagator sign is evidence, but does not by itself decide whether the theory contains a physical ghost particle.

Four logically distinct objects must not be conflated:

1. solutions of the local field equations;
2. structures remaining after gauge, charge, constraint, and boundary reduction;
3. directions that survive nonlinear continuation and interactions;
4. quantum states and asymptotic particles with a physical probability rule.

Substantial parts of the first two layers are now exact. Complete free causal gauge complexes have been constructed on several declared backgrounds. On a compact Einstein–Maxwell/Weyl–Maxwell laboratory, the Einstein image and additional fourth-order axial and polar summands are explicit. The axial additional block is nonradical under the action-derived Lee–Wald form, so it cannot there be dismissed as determinant multiplicity or local gauge. The sign distribution also does not coincide with the simple slogan “Einstein healthy, extra ghostly.” These are classical current results, not quantum norms.

The nonlinear answer is sector-dependent. In the certified finite harmonic space, a finite exponential-polynomial second-order correction exists exactly on the zero set of five stabilizer moment maps—global quadratic charge-balance conditions. Bounded quasiperiodic continuation obeys further polynomial-growth and resonant-shell conditions. Some pure additional directions are therefore obstructed, while balanced combinations can possess formal secular extensions. Formal continuation is not stability.

A changed classical theory has now passed a new constructive gate. Two positive phase fields share one auxiliary diagonal  $U(1)$  gauge connection; their gauge-neutral relative phase supplies a clock while the diagonal phase is contractible. On one selected fixed- $Q_{\text{rel}}$  Berger background, the action is exactly equivalent to the previously certified positive Berger clock system plus a contractible gauge summand. This gives a complete real cyclic 70-component BV parent with support-local advanced and retarded chain homotopies. It is the first compensator-successor model to pass the complete classical causal-parent gate. Its physical cohomology, reduced inertia, interactions, observables and quantization were therefore tested as separate promotion gates rather than inferred from causality. An exact homogeneous stationary classification has also shown that the passing fixed-action point is isolated rather than an open phase, and that the same action has no stationary path to the round cylinder. The homogeneous trace block is positive and oscillatory there; this does not replace the full nonhomogeneous physical reduction. That fixed-charge reduction has now been performed in the relative global sector: fixing  $Q_{\text{rel}}$  and quotienting its shift removes the entire relative-clock Darboux pair. Thus the branch on which  $D$  becomes null contains no physical relative clock. The alternative unrestricted branch retains the clock pair, but raw  $D$  is then a charged physical symmetry. Its charge curvature is positive and it has no real exponentially growing root. Its exact size-two zero Jordan block is now identified as the tangent to an integrable action-angle family: changing the relative charge changes the clock rate and therefore produces a phase drift linear in time. The augmented energy has one positive transverse direction, no negative direction, and the physical phase orbit as its zero direction. Fixed-charge orbital stability and shifted-frequency modulated stability pass, although absolute stability of a lifted or compact phase fails when different clock rates are compared. This reduced family is not a nearby family of complete coupled Berger backgrounds: at the selected geometry the full constraints force the charge variation to vanish. The first attempted all-Hodge carrier now fails by an exact representation-theoretic mechanism: on the squashed Berger sphere, the round-style scalar exact-one-form sector is not closed under the gauge-fixed differential. Its leading defect is  $93ik/40$  and is nonzero for every right weight  $k \neq 0$ ; the round-space mutation removes it. This is a failure of that decomposition, not a negative spectrum or a defect of the causal 70-row parent. The replacement closes complete  $SU(2)_L \times U(1)_R$  isotypical blocks and forms their physical quotients for every  $m, k$  at  $j = 0, \frac{1}{2}, 1$ . Every certified quotient is nonradical but physically unstable; for  $j \geq \frac{3}{2}$  the physical quotient and spectrum remain NO\_CERTIFIED\_MAP. The exact same-field retuning theorem then closes the candidate-selection question without extrapolating to those higher isotypes: the complete  $j = \frac{1}{2}$  quotient retains its Hamiltonian-Hopf quartet throughout the connected trace-healthy stationary family. There is therefore no open or structurally protected linearly healthy Phase-2 candidate in that declared family. This leaves the selected causal parent intact and is not a no-go for changed field content, derivative order, or action architecture.

Two further promotions have now failed cleanly at their first maps. The selected

Berger counterflow background is not a solution of conventional Einstein gravity coupled to the same matter for any constants  $(\Lambda, \kappa)$ , so there is no same-base-point Einstein tangent inclusion or additional-mode cofiber to construct there. Separately, the background’s five-dimensional stabilizer algebra is exact, but the 70-row parent does not yet export the four spatial symmetry actions and moment maps needed for a residual BFV receiver. These are scoped incidence and carrier obstructions, not failures of the causal parent or universal no-go theorems.

At one Euclidean loop order, strict fixed-field-content pure Weyl gravity has a nonzero local BV anomaly. A tau-adic compensator extension makes that local anomaly exact and restores the Euclidean quantum master equation at the tested order, but it enlarges the BV complex and counterterm space: it is a changed theory, not proof that strict pure Weyl gravity is anomaly-free. A separate reduced free mode space admits Hadamard two-point functions with signs  $(+, -, -)$ . No positive full-BV state, asymptotic particle space, scattering theory, or unitarity theorem presently exists.

The current verdict is therefore neither “ghost removed” nor “theory ruled out.” Additional classical branches survive several gauge, radical, and horizon tests, while both parities at one Schwarzschild fixture are removed by finite symplectic norm at infinity. Nonlinear and quantum calculations impose further real obstructions; the general asymptotic and full-BV physical quotients remain open.

**Phase 1 is closed.** Its outcome is a classification rather than a selected theory: the programme constructed its first causal changed-theory candidate, found a nonradical Hamiltonian–Hopf instability that persists across the connected trace-healthy same-field family, and did not activate interaction, observer, or quantum promotion for that candidate. This does not rule out a genuinely different action or field content in a future Phase 2.

## Current verdict

| Established                                                            | Partial                                                                 | Open                                   |
|------------------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------|
| Complete causal free complexes on several declared backgrounds         | Relational clocks, causal redshift, and localized detector preparations | Invariant branch-resolved interactions |
| Explicit Einstein-image and additional fourth-order classical summands | Bounded nonlinear continuation on selected finite mode spaces           | Complete bounded finite-support cone   |

| Established                                                                                                                                                                                                                                            | Partial                                                                                                                                                                                                                    | Open                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Nonzero Lee–Wald pairings for selected additional directions                                                                                                                                                                                           | Schwarzschild horizon admissibility plus two-parity finite-norm Einstein selection at one fixture                                                                                                                          | Asymptotic Bach phase space and scattering                                                                                                          |
| Exact nonlinear balance conditions, a repaired- $q_{70}$ health obstruction on every certified $j = 0, \frac{1}{2}, 1$ physical block, the familywide $j = \frac{1}{2}$ same-field retuning obstruction, and a strict local Euclidean one-loop anomaly | A two-phase counterflow changed theory with a certified selected-fixture causal BV parent and exact charge–clock complementarity; compensator restoration and reduced Hadamard/Krein two-point functions on other carriers | Higher- $j$ counterflow spectra, changed-action architectures beyond the declared family, a full-BV state, Lorentzian QME, particles, and unitarity |

The public construction map shows the dependency structure; the searchable residual atlas tracks individual mode families without identifying them across backgrounds.

## Starting choices and distinct laboratories

The initial cylinder calculation makes choices, not conclusions:

| Starting choice                              | Purpose                                                               | It does not imply                                                |
|----------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------|
| Strict four-dimensional pure-Weyl action     | Stress-test locally conformal fourth-order gravity                    | Correctness, stability, unitarity, or phenomenological viability |
| Conformal cylinder $\mathbb{R} \times S^3$   | An exact closed causal laboratory for the complete free gauge complex | The topology or boundary conditions of the observed universe     |
| Free linearized vacuum theory                | Separate propagation and reduction from interaction failures          | Nonlinear closure, matter compatibility, particles, or cosmology |
| Diffeomorphism and local Weyl gauge symmetry | Remove the defining local redundancies                                | That every residual conformal generator is proper gauge          |

| Starting choice                     | Purpose                                                                                     | It does not imply                                       |
|-------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Selected zero-charge derived sector | Test the proposed residual quotient after deriving its charge condition                     | That cylinder time translation $D$ is universally gauge |
| Full BV–BFV gauge complex           | Track fields, equations, ghosts, identities, constraints, boundaries, and pairings together | That a raw pole or local mode is a physical state       |

Here **BV–BFV** means the gauge-complex framework containing fields, equations, ghosts, constraints, and boundary data. A **residual quotient** is the further quotient by gauge symmetries left after local gauge fixing.

Different questions require different backgrounds:

| Laboratory                                        | Principal test                                                                                                     |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Conformal cylinder $\mathbb{R} \times S^3$        | Complete free causal complex and selected residual reduction                                                       |
| Berger clock backgrounds                          | Clocks, relational observables, gravity–matter interactions, and the two-phase counterflow causal-parent candidate |
| $\mathbb{R} \times S^1 \times S^2$                | Einstein/additional waves, Taub constraints, and resonances                                                        |
| Nariai                                            | Causal transfer on a different curved background                                                                   |
| Schwarzschild and the static spherical family     | Horizons, thermodynamics, and black-hole perturbations                                                             |
| Euclidean $S^4$ and generic Euclidean mode spaces | Anomaly, Hessian, and determinant calculations                                                                     |

**These are distinct laboratories. A result on one background does not automatically transfer to another.** Every comparison is indexed by

(theory, background, generator, phase space, boundaries, lifecycle).

## Four flagship results

### A. Causal gauge systems and residual reduction

On the conformal cylinder, the complete free metric BV–BFV complex has exact retarded and advanced Green homotopies, transport between support conditions, and matching Green, current, and residual pairings. Thus the residual calculation

is connected to the covariant field theory rather than being an isolated Fourier quotient.

In the selected closed, zero-charge derived sector, all fifteen residual  $SO(4, 2)$  generators are imposed as constraints. The vacuum and selected residual one-particle cohomology are acyclic. The first surviving classes are

$$H^4 = \text{span}\{[W_+^2], [W_-^2]\}, \quad G = I_2.$$

They are deformation or vertex classes, not graviton particles. The sector qualification is essential:  $D$  is charged on the unrestricted compact phase space and becomes gauge only on the Taub-zero derived sector. A Taub obstruction is a second-order global integrability obstruction; its moment-map form is a quadratic charge-balance condition.

The finite residual mode window is not an arbitrary-support replacement for the all-energy causal theory. An exact weight-five test finds 64 Einstein cohomology directions in the source and none in the selected target, so a  $D$ -equivariant unary deformation retract to that finite receiver cannot exist. This leaves the finite residual theorem intact, but blocks the proposed arbitrary-support transfer of the nonlinear operation and anomaly classes until an all-energy completed time-slice carrier is constructed.

The causal-transfer theorem also has Berger, flat-Minkowski, and curved Nariai consumers. On Nariai the complete repaired system retracts to the metric Bach endpoint with exact causal homotopies and one certified formal transverse background variation.

The newest Berger result concerns a genuinely changed action rather than a passive compensator. Two clock phases counterflow so that the diagonal compact gauge charge vanishes while the relative phase evolves. On the selected squashed background and fixed relative-charge leaf, the resulting theory has a complete 70-component action-derived causal BV parent. The exact fixed-action homogeneous stationary locus contains this positive solution as an isolated point and supplies no stationary route to the round cylinder. Raw  $D$  remains charged on the unrestricted parent. On the derived fixed- $Q_{\text{rel}}$  fibre, quotienting the relative shift makes  $D$  null but also contracts the complete relative-clock Darboux pair: the physical clock dimension is zero. On the distinct unrestricted branch the clock survives and time translation is a charged physical symmetry. Its exact homogeneous charge block has positive curvature and no exponential root. The size-two zero Jordan chain is exactly the tangent to an integrable action-angle family with charge-dependent clock rate, not a negative-energy mode. It is fixed-charge orbitally stable and shifted-frequency modulated stable, while absolute comparison of clocks with different rates accumulates phase. The complete coupled background is nevertheless locally isolated and does not inherit that nearby reduced family. The round-style scalar Hodge carrier is now exactly obstructed for  $k \neq 0$ ; the next Classical gate is a full closed Berger

isotypical carrier before any pairing, gradient, characteristic or radical census. The homogeneous normal form is not a full health theorem.

**Immediate limitation.** This is not a universal removal of radiative gravitons, an open viable phase for generic backgrounds, or an asymptotic particle theorem.

## B. Einstein-image and additional classical branches

On  $\mathbb{R} \times S^1 \times S^2$ , the complete linear Einstein–Maxwell tangent maps into the Weyl–Maxwell system. The resulting linear triangle and its relative space are explicit, including global and large-gauge endpoints. The triangle is not cyclic under the standard pairings: Einstein inclusion is not symplectic equivalence.

For generic axial  $\ell \geq 2$ , quotienting the Weyl–Maxwell solution space by the Einstein–Maxwell image leaves two exact additional polarizations. A direct four-dimensional Lee–Wald calculation makes this additional block nonradical with signature  $(2, 0)$ ; the complete generic axial space has signature  $(3, 1)$ , with the negative direction on an Einstein-image master branch. “Einstein” and “positive,” or “additional” and “negative,” are therefore not synonymous on this compact classical space.

The polar operator independently contains the complete Einstein primary summand plus two additional summands on every declared physical fibre. A literal four-dimensional curvature-momentum calculation, including the load-bearing variation of  $\nabla C$ , now reproduces the reduced current exactly for generic  $\ell \geq 2$  and compact momentum. The two additional polar directions are nonradical with inertia  $(2, 0)$ ; the Einstein-image block has inertia  $(1, 1)$  and the complete polar target  $(3, 1)$ . The axial and polar blocks therefore have the same inertias without identifying their representatives across parity.

The corresponding ungauged equation/Noether chain map exists, but its cyclic promotion now has a stronger invariant obstruction. With the frozen identity field inclusion, the polar cyclic defect has rank two on every generic physical fibre. Changing the extra lift can change raw cross terms and even a chosen lifted-extra sign, but it cannot change the Schur-complement quotient form. More generally, the Einstein–Maxwell source form has inertia  $(2, 0)$  while the Weyl–Maxwell restriction to its Einstein image has inertia  $(1, 1)$  in both generic parities. No time-translation- and shell-preserving complex-linear correction can identify them cyclically. What survives is a canonical target-internal orthogonal complement and a parity-complete short exact sequence on solution cohomology before global reduction: generic fibres have dimensions  $4 \rightarrow 8 \rightarrow 4$ , zero radicals, and a noncyclic mapping-cofiber triangle at all-row level.

**Immediate limitation.** Every strict cyclic split in the declared parity-complete, time-translation-preserving class is obstructed. A non-strict cyclic homotopy can only be posed after constructing the missing common moment-map-zero derived carrier. Final residual de-

scent, causal boundaries and positive-frequency quantization remain open. A nonzero classical pairing is not a particle norm.

### C. Nonlinear continuation, interactions, and resonance

Paper 13 classifies the second-order tangent problem on the certified finite harmonic Weyl–Maxwell space. A **finite exponential-polynomial correction** is a finite sum of  $t^p e^{-i\Omega t}$  terms. Such a correction exists exactly when five stabilizer moment maps vanish. This is an exhaustive model-specific realization of the classical linearization-stability and moment-map framework, not a claim to have invented that general theory.

Bounded quasiperiodic continuation is stricter. It additionally requires the vanishing of positive polynomial-growth coefficients and adjoint pairings on resonant characteristic shells. On the complete global plus ( $\ell = 2, k = 0$ ) additional-primary space, the bounded cone reduces to the standard global generalized-zero cone: no nonzero additional-primary amplitude survives. Other balanced finite combinations can possess formal exponential-polynomial corrections with secular growth. The full bounded finite-support zero locus remains open.

The standard-branch calculation now extends uniformly to every  $\ell \geq 2$  at a separately tuned compact momentum. Its exact resonance zero set consists of two mixed axial–polar sheets and two one-sided planes; after the global energy and momentum balances, the bounded cone is the origin plus the two mixed sheets on an explicit occupation interval. In a separate two-absolute-momentum enlargement, all 108 axisymmetric  $L = 4$  and all 56 nonaxisymmetric  $L = 1, 3$  branch-basis coefficients are exact and individually obstructed. This complete 164-coefficient basis census is not yet the arbitrary-amplitude zero variety: cross-column cancellations remain open.

The Berger interaction programme has exact cyclic  $q_2, q_3$  tensors and a retained mixed gravity–Maxwell  $\ell_3$  representative on its declared mode space. That representative is an architectural result, not yet an invariant physical interaction. Once the fuller Maxwell ghost structure is included, the physical-action mixed interaction is cyclically removable through summed input-jet order two. On the complete full-BV bounded cyclic complex, the former second-jet witness is invalid but no full primitive or replacement witness is certified. The invariant class and every Einstein/additional/Maxwell mixing table therefore remain open. The nonlinear symmetry that does close is  $K_{\text{Berger}} = D - \omega R$ , not raw  $D$ .

The Einstein–Maxwell-to-Weyl–Maxwell inclusion has also reached its first nonlinear obstruction. The common action-derived  $q_1, q_2, q_3$  tensors pass their internal identities, but the strict arity-two inclusion defect is nonzero. Taub pairing rules out a support-local correcting  $f_2$  on the full smooth periodic fixed-bundle carrier. Its global cohomological image is nevertheless exactly the five stabilizer charges. Locality therefore forces the relative receiver to be a Noether-current complex, with charges obtained only after integrating over a Cauchy slice. The canonical Hessian Green current and its local divergence identity are now exact



on all fourteen physical rows. Stabilizer precomposition, cyclic-dual completion, Lee–Wald comparison, and recovery of all five integrated charges are the next gates.

**Immediate limitation.** Exponential-polynomial continuation is not bounded stability, retarded solvability, Sobolev existence, an invariant interaction amplitude, or all-order nonlinear integration.

#### D. Quantum anomaly and black-hole endpoint tests

For strict fixed-field-content pure Weyl gravity, the classified local one-loop Euclidean BV anomaly has nonzero coefficients. The local Euclidean quantum master equation is therefore obstructed. In a tau-adic compensator extension, the anomaly becomes exact and the local equation is restored at that order. This second result changes the BV complex and counterterm space; it does not make strict pure Weyl gravity anomaly-free.

Separately, a reduced free cylinder mode space with Einstein-like, additional, and longitudinal families admits compatible Hadamard two-point functions—the required short-distance singularity structure—with indefinite signs  $(+, -, -)$ . These Krein signs are signs in a candidate state pairing, not a full BRST-positive Hilbert space. The free curvature-observable CCR algebra is also exact, but it has no certified compatible positive state. On  $S^2 \times S^2$ , a separate Euclidean spectral calculation now gives the exact longitudinal ghost Schur eigenvalue and a coupled exceptional-mode correction. It advances the determinant assembly but remains EUCLIDEAN-SPECTRAL, not a Lorentzian state or unitarity result.

The black-hole programme classifies the static spherical Bach-flat family and obtains a consistently normalized energy, Wald entropy, and first law on its declared domain. On Schwarzschild, the axial Einstein/Regge–Wheeler family and an additional Ricci-curvature family are explicit. The additional family has future-horizon-regular ingoing solutions. Controlled real-frequency fixtures give nonzero Einstein–additional and additional–additional horizon pairing, while the pure Einstein block is null in the pure-Weyl current.

Horizon analyticity alone does not force the additional family to vanish. But at  $\ell = 2$ ,  $\omega = 3/5$ , the completed axial and polar metric reconstructions have divergent Einstein–extra and extra–extra slice norms, while the Einstein sector is finite (with an additional zero polar self-density in one sector). Thus finite symplectic norm at infinity selects exactly the Einstein sector in both parities at this fixture. This is a global phase-space selection, not a local horizon condition or a general all-frequency theorem.

**Immediate limitation.** A symbolic-frequency theorem, general  $\ell$ , summability, a complete asymptotically flat phase space, the exterior initial-boundary problem, complex frequencies, ringdown, and stability remain open.

## Where the strongest criticism currently lands

The strongest criticism is correct on six points:

- $D$  is not universally gauge; it is charged on the unrestricted compact phase space and is expected to be physical in ordinary asymptotic settings.
- Acyclic selected cylinder one-particle cohomology is not asymptotic unitarity or absence of radiative degrees of freedom.
- Einstein inclusion is not nonlinear closure, symplectic equivalence, or an exclusive sector of Weyl gravity.
- Additional classical branches survive selected gauge-radical and horizon tests, while finite symplectic norm excludes the extra branch in both parities at one certified Schwarzschild fixture.
- The Berger nonlinear generator is  $K_{\text{Berger}} = D - \omega R$ , not raw  $D$ , and the retained interaction has no certified invariant branch meaning.
- The compensator repairs a changed local Euclidean theory; it does not remove the strict theory's anomaly.

What remains valid is correspondingly precise: the zero-charge condition is derived rather than assumed; the selected residual quotient is connected to the covariant causal complex; a healthy clock can coexist with total zero-charge  $D$  at linear fixed coupling; the compact additional axial space is nonradical; exact nonlinear balance and resonance obstructions exist; and the strict/extended quantum alternatives are computationally separated. The two-phase counterflow construction supplies one changed-theory causal parent with an exact charge-clock tradeoff and homogeneous action-angle stability theorem. Its first round-style scalar Hodge carrier is exactly not a subcomplex on the anisotropic Berger background. Complete isotypical closures replace that invalid split for  $j = 0, \frac{1}{2}, 1$ , and every certified physical quotient is nonradical but unstable. The  $j = \frac{1}{2}$  quartet then persists over the connected trace-healthy same-field retuning family, so this candidate is terminally not selected for Phase 2 without claiming a complete higher-isotype spectrum or a universal changed-theory no-go.

## Authoritative result ledger

The rows below are the authoritative short formulations. Raw implementation milestones are excluded.

| Scientific question                         | Current result                                                                                                                                                                                                                                         | Lifecycle                                                                                                                               | Principal limitation / route                                                                                                                                                                   |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Free causal complex and residual cohomology | Complete cylinder causal complex; selected zero-charge residual vacuum and one-particle cohomology acyclic; $H^4 \cong \mathbb{C}^2$ contains deformation classes                                                                                      | <b>Certified on their distinct carriers</b>                                                                                             | The finite residual receiver is not an arbitrary-support all-energy retract; the latter has a rank-64 unary obstruction and requires completion; Papers 7–8                                    |
| Is $D$ gauge?                               | Charged on the unrestricted compact phase space; gauge on the Taub-zero derived sector                                                                                                                                                                 | <b>Certified, sector-dependent</b>                                                                                                      | Must be recomputed with boundaries and matter; status ledger                                                                                                                                   |
| Clocks and redshift                         | Healthy original Berger clock and a spatially global retarded $G_0$ probe fixture with $1 + z = 2$ on its declared carrier; the changed theory has a charged action-angle clock plus exact distributional and finite-resolution conditional event maps | <b>Probe-mode fixture / changed-theory clock and conditional sampling theorems certified; operational apparatus ratio not activated</b> | The standalone local receiver cocycle cannot be included by a degree-zero map into repaired $q_{70}$ ; no residual nonradical receiver or localized operational redshift is certified; Paper 9 |

| Scientific question                         | Current result                                                                                                                                                                                                                        | Lifecycle                                                                                            | Principal limitation / route                                                                                                        |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Einstein image and additional branches      | Linear inclusion, parity-complete pre-residual $H^0$ exact sequence and nonradical currents certified; lift-shear invariants give a canonical target-orthogonal split but obstruct every declared parity-complete strict cyclic split | <b>Pre-residual exact sequence / direct currents / invariant strict-cyclic obstruction certified</b> | A common charge-zero derived carrier, any admissible non-strict homotopy, causal boundaries, and quantization remain open; Paper 92 |
| Nonlinear tangent cone and resonances       | Formal finite exponential-polynomial cone equals five-charge zero set; tuned standard bounded cone is classified for every $\ell \geq 2$ ; all 164 two-momentum basis coefficients are closed                                         | <b>Formal cone and all-<math>\ell</math> standard subcone certified; larger cone partial</b>         | Arbitrary-amplitude two-momentum zero variety, causal, infinite-mode, and all-order problems open; Paper 13                         |
| Interaction and branch mixing               | Retained mixed tensor is exact/cyclic; the physical action is trivial through summed input-jet order two                                                                                                                              | <b>Representative certified; complete full-BV invariant class open</b>                               | A full-BV primitive or replacement cokernel witness and branch-resolved operation are not certified; Paper 11                       |
| Free quantum observables and reduced states | Curvature CCR algebra exact; separate reduced E/A/L Hadamard carrier has signs $(+, -, -)$                                                                                                                                            | <b>Certified on distinct free carriers</b>                                                           | No full-BV positive state, particles, or interacting products; state report                                                         |

| Scientific question                | Current result                                                                                                                                                                                | Lifecycle                                               | Principal limitation / route                                                                             |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| One-loop anomaly                   | Strict local Euclidean QME obstructed; compensator extension restores changed theory at one loop                                                                                              | <b>Strict obstructed / extension restored</b>           | No Lorentzian QME or unitarity theorem; Paper 12                                                         |
| Black holes                        | Static first law; additional axial and polar families reach the horizon, but finite symplectic norm at infinity selects the Einstein sector at the certified $\ell = 2, \omega = 3/5$ fixture | <b>Static certified / two-parity fixture classified</b> | Symbolic frequency, general $\ell$ , full scattering phase space, stability, and ringdown open; Paper 14 |
| Asymptotic particles and unitarity | No theorem                                                                                                                                                                                    | <b>Open</b>                                             | Requires asymptotic Bach phase space, positive physical state, scattering, and optical/unitarity tests   |

## Highlights by audience

### Higher-derivative gravity and unitarity

The compact current calculation separates extra roots from gauge and radical directions, while placing one negative classical direction on an Einstein image rather than the additional axial block. A potentially distinctive element is the common comparison of local modes, charge sectors, nonlinear obstructions, and reduced quantum signs. The decisive gate is a positive full-BV or asymptotic physical state, not another pole count.

### BV/BRST and perturbative quantum field theory

The programme supplies complete classical gauge-complex inputs, a free causal curvature algebra, a classified one-loop local anomaly, and a compensator ex-

tension in which that anomaly is exact. A potentially distinctive element is the fail-closed separation of local algebraic, Euclidean spectral, reduced-mode, and Lorentzian causal claims. The open gate is a renormalized Lorentzian QME and a positive state on the full BV complex.

### **Lorentzian PDE and Green-hyperbolic complexes**

Exact retarded/advanced Green homotopies transfer through cyclic differential contractions to fourth-order gauge endpoints on several backgrounds. A potentially distinctive element is constructive pairing transport through tractor/HPL reductions when no isolated central operator provides the whole causal object. The open gate is stability on a finite family of less symmetric globally hyperbolic backgrounds.

### **Conformal geometry and detour complexes**

The tractor parent, BGG/detour endpoint, differential contraction, and Lorentzian Green homotopy are explicit in one large conformal-gravity example. A potentially distinctive element is the causal interpretation of the Bach operator as a compression of a larger gauge complex. The next gate is one second detour or higher-spin consumer and a background-family theorem.

### **Mathematical relativity and black holes**

The compact programme gives explicit charge-fibre-dependent Taub obstructions and all- $\ell$  resonant refinements; the black-hole programme gives a static first law, Ricci-carrier factorization, horizon reach, and finite-norm Einstein selection in both parities at one fixture. Potentially distinctive elements are the exhaustive finite-mode cone and the pure-Weyl horizon/asymptotic pairing comparison. The gates are the joined two-momentum zero variety and a complete Schwarzschild exterior phase space.

### **Relational observables and clocks**

A healthy Berger matter clock coexists with zero total  $D$  charge in one linear fixed-coupling sector, and a retarded Maxwell signal defines a gauge-invariant spatially global clock-slice  $G_0$  probe observable with  $1 + z = 2$  on that declared carrier. The complete receiver audit preserves this probe theorem but does not promote it to a localized action-derived physical receiver or operational apparatus redshift. The standalone local receiver cocycle exists; its degree-zero chain/cochain inclusion into repaired  $q_{70}$  is the exact Phase-1 obstruction.

The changed two-phase route has a different and now exact charge-clock geometry. Fixing and reducing the relative charge makes  $D$  null but removes the clock; retaining the clock makes  $D$  charged. On the unrestricted branch the homogeneous clock is an integrable action-angle system with positive transverse energy and charge-dependent rate. A certified event-map preflight constructs

a local-gauge-closed clock-slice observable and proves covariance, rather than quotient invariance, under the charged  $R_{\text{rel}}$  and  $D$  flows. The repaired- $q_{70}$  health assembly subsequently forms the physical quotients for  $j = 0, \frac{1}{2}, 1$ : every certified block is nonradical but unstable, while the  $j \geq \frac{3}{2}$  quotient and pairing are undefined. Fixed charge removes the clock. On the unrestricted connected trace-healthy same-field family, the exact  $j = \frac{1}{2}$  discriminant remains negative and the Hamiltonian–Hopf quartet persists. Consequently no robust Phase-2 candidate, healthy common physical clock–receiver carrier, thirteen-field receiver interface or counterflow redshift is activated. The higher-isotype retuning spectrum remains `NO_CERTIFIED_MAP`; it is not needed for this scoped nonselection and is not silently promoted.

This preflight has also been promoted to a general exact theorem for any physical action–angle clock with charged time translation. It proves the distributional event map, its local BV descent, primitive clock-basis composition and phase-origin covariance, while separating lifted-phase boundedness from compact orbital and modulated stability. Nontriviality is explicitly conditional on a surviving receiver cohomology class and pairing. An exact finite-resolution version replaces the sharp clock slice by an even compact sampling profile. It preserves local BV closure and causal support, converges distributionally, and gives a quadratic clock-width error bound  $\epsilon^2 \mu_2 \sup |P''|/2$ . Smoothing does not create the still-missing receiver class, so this is not yet a counterflow redshift measurement. Conditional nonzero sampled records also obey exact reciprocity and composition laws: adjacent ratios compose only when the intermediate charge fibre and sampled record are identified, while a closed comparison loop measures the product of its fibre/orientation transitions. An algebraic inverse is not an advanced physical signal, and unmatched fibres give no map rather than an anomalous measurement.

## What would materially change the verdict

A major positive shift would be produced by any one of: a nondegenerate positive asymptotic Einstein sector with the additional family excluded by a physical boundary rule; a positive full-BV Hadamard state; a certified nontrivial branch-resolved interaction compatible with the constraints; or a localized backreacting relational measurement.

A major negative shift would be produced by: an unavoidable negative physical direction after residual and boundary reduction; a nontrivial Lorentzian quantum Cartan anomaly; failure of every admissible nonlinear Einstein-sector boundary condition; or an asymptotic additional branch with negative physical flux that cannot be consistently excluded.

Either outcome is valuable. The programme is designed to locate the first precise obstruction, not to protect a preferred interpretation.

## What collaborators can use now

The repository can support a collaborator who wants to:

- audit one flagship theorem against an independent convention;
- supply a preferred operator, gauge, boundary condition, or background;
- compare a known result with a complete certificate chain;
- inspect one unresolved defect or deliberately broken verifier input;
- help construct the asymptotic Bach phase space;
- review the anomaly/BV argument or the nonlinear tangent-cone theorem;
- run an adjacent detour, higher-spin, black-hole, or observer problem through the existing symbolic infrastructure.

The programme is not asking a new collaborator to endorse the entire interpretation. It is asking them to attack one scoped claim with full access to its derivation and verifier.

## Detailed active projects

| Project                        | Current boundary                                                                                                                                                                | Exact next gate                                                                                                                                    |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Cyclic causal Green transfer   | Abstract conditional theorem with Berger, flat, and Nariai consumers; one Nariai formal tangent                                                                                 | Extend to a finite background family or second detour theory.                                                                                      |
| Charge-fibre obstruction       | Formal cone, all- $\ell$ tuned standard cone, and all 164 two-momentum basis coefficients certified                                                                             | Classify the common amplitude zero variety.                                                                                                        |
| Additional axial/polar current | Direct currents and the maximal pre-residual $H^0$ exact sequence are parity complete; the invariant source/image inertia mismatch obstructs every declared strict cyclic split | Construct the common constraint-algebroid/Kuranishi moment-map-zero carrier, then ask whether a non-strict homotopy and physical boundaries exist. |



| Project                       | Current boundary                                                                                                                                                                                                                                                                                                 | Exact next gate                                                                                                                                                                                    |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Relational clock and light    | Original spatially global retarded $G_0$ probe fixture certified on its declared carrier; the changed theory has exact conditional comparison laws, but its repaired- $q_{70}$ certified physical blocks are unstable and the $j = \frac{1}{2}$ quartet persists across the connected same-field retuning family | The operational counterflow ratio and Phase-2 candidate are <b>NOT_ACTIVATED</b> ; a new action architecture would require its own causal, health, receiver and $D/R_{\text{rel}}/K$ certificates. |
| Black-hole radiation          | Static theorem and horizon reach in both parities; finite norm at infinity selects Einstein in both parities at one fixture                                                                                                                                                                                      | Generalize in frequency and $\ell$ , then construct the asymptotic phase space.                                                                                                                    |
| Einstein–Weyl relative theory | Matching action-derived tensors through $q_3$ ; strict $f_1$ has a nonzero arity-two defect; global five-charge receiver and Hessian Green-current cone certified                                                                                                                                                | Precompose all stabilizers, add cyclic duals, compare with Lee–Wald, and determine the correct relative nonlinear extension.                                                                       |
| Residual branch mixing        | Retained cyclic representative and rank-46 carrier exist; physical-action order two is trivial, while the complete full-BV interaction class is open                                                                                                                                                             | Decide the full-BV bounded cyclic class and obtain an admissible branch resolution.                                                                                                                |
| Quantum anomaly and states    | Strict anomaly, compensator restoration, reduced Hadamard carrier, and determinant subproblems certified                                                                                                                                                                                                         | Assemble the physical Hessian/determinant and test the Lorentzian full-BV QME/state.                                                                                                               |
| Asymptotic Bach/BMS           | Programme stage only                                                                                                                                                                                                                                                                                             | Construct differentiable charges, flux, falloffs, and signs at null infinity.                                                                                                                      |

## Method and computational accountability

Palm sets the scientific direction and retains editorial responsibility. AI systems propose and execute much of the detailed mathematics, software, review, literature comparison, and drafting. Neither credentials nor fluent output count as evidence. The certified systems range from dozens to hundreds of linked field and gauge components and can contain more than one hundred thousand exact symbolic interaction terms.

Every material claim is expected to provide: a typed scope; exact or controlled inputs; a machine-readable certificate with hashes and unresolved fields; an independent verifier; mutation tests that must fail on damaged inputs; and a human-readable statement of what was and was not established. Convention repairs and withdrawn promotions remain visible. Publication artifacts are rebuilt from pinned sources. This makes the repository a symbolic laboratory rather than a collection of unchecked AI prose.

Technical manuscripts, certificates, and verification reports

### Core papers

- Series overview
- Paper 7: residual cohomology
- Paper 8: covariant causal transport
- Papers 7–8 computational supplement
- Paper 9: relational clocks
- Paper 10: compact Einstein–Maxwell/Weyl–Maxwell phase space
- Paper 11: retained mixed gravity–Maxwell tensor
- Paper 12: one-loop BV anomaly
- Paper 13: finite-harmonic second-order tangent cone
- Paper 14: pure-Weyl black-hole radiation

### Causal geometry

- Cyclic Green-transfer bridge
- Curved Nariai causal transfer
- Formal transverse Nariai variation
- Free curvature-observable causal propagator
- Free curvature CCR algebra

### Einstein/additional comparison

- Additional-branch current bridge
- Generic axial operator report
- Polar operator certificate
- Polar physical module certificate
- Complete noncyclic linear triangle
- Einstein–Maxwell tensors through arity three

- Relative arity-two defect
- Relative five-charge receiver
- Why the local receiver must be a current
- Polarized relative current seed
- Relative Hessian Green-current cone

### **Nonlinear tangent cone and interactions**

- Charge-fibre bridge
- Complete global plus ell2 bounded cone
- Standard global bounded cone
- Aligned additional bounded obstruction
- Aligned formal secular extension
- All-ell tuned standard cone
- Two-momentum axisymmetric workload
- Two-momentum nonaxisymmetric L3 matrix
- Two-momentum nonaxisymmetric L1/L3 completion
- Berger first-jet redefinition
- Berger second-jet source
- Rank-46 cyclic branch carrier

### **Clocks and observers**

- Retarded spatially global Maxwell probe observable
- Localized emitter rank-two response
- Mass-uniform leading response
- First retarded emitter channel
- 108-row first-jet unary obstruction

### **Black holes**

- Static spherical family
- Normalized generator and first law
- Axial operator
- Axial horizon reach
- Axial horizon pairing
- Scoped axial endpoint analysis
- Axial finite-norm boundary class
- Polar operator
- Polar horizon reach
- Polar finite-norm boundary class

### **Quantum anomaly and states**

- Anomaly coefficients and D descent
- Anomaly-induced action
- Reduced cylinder Hadamard/Krein carrier

- Berger microlocal gate
- Generic ghost Schur reduction
- Product  $S^2 \times S^2$  ghost Schur carrier

## Release and audit infrastructure

- Live D-quotient ledger
- Residual atlas
- Technical certificate graph
- Publication release audit
- Universe-building roadmap
- General-audience introduction

## Detailed changelog

- **21 July 2026:** froze the Classical Phase-1 counterflow chain: retained the selected-fixture causal parent, recorded that fixed relative charge removes the clock, and propagated the familywide  $j = \frac{1}{2}$  Hamiltonian–Hopf obstruction and terminal same-field nonselection without extrapolating to higher isotypes or other action architectures.
- **21 July 2026:** preserved the spatially global retarded  $G_0$  probe observable on its declared carrier, removed its unsupported promotion to a localized operational apparatus redshift, and recorded the exact degree-zero receiver chain/cochain inclusion obstruction as the Phase~1 endpoint.
- **19 July 2026:** revised Paper 13 terminology to finite exponential-polynomial continuation; added the complete global plus ( $\ell = 2, k = 0$ ) bounded subcone and retained the open full bounded cone. Narrowed the Schwarzschild result to horizon-and-leading-asymptotic nonselection, added polar horizon reach, and updated observer and one-loop determinant work. Corrected the Berger mixed-interaction status to removal through physical-action input order two while keeping the complete full-BV bounded cyclic class open.
- **18 July 2026:** added the noncyclic Einstein–Weyl linear triangle, Einstein–Maxwell tensors through  $q_3$ , the formal Nariai causal variation, black-hole horizon pairing fixtures, reduced cylinder Hadamard/Krein signs, and the first bounded/secular nonlinear classifications.
- **17 July 2026:** added the retarded Berger redshift observable, mixed  $q_3$ , retained  $\ell_3$ , rank-46 carrier, localized detector preparations, and the corrected nonlinear generator  $K_{\text{Berger}} = D - \omega R$ .
- **16 July 2026:** created the physicist-facing summary; recorded the cylinder causal bridge, sector-dependent  $D$ , generic axial strict inclusion, and fail-closed Berger microlocal promotion.

Historical entries record what was known at the time. The scientific sections and result ledger above are authoritative when an older phrase differs.

## Update policy

This remains the programme's single physicist front door. A new result changes the visible narrative only when a lifecycle advances or retreats, a physical dependency changes, a new background or boundary changes a verdict, a decisive open test closes, or a manuscript changes publication state. Raw term counts and implementation progress remain in team reports.

Each update must preserve one authoritative wording per result, place its strongest limitation beside it, keep distinct backgrounds separate, and retain the technical evidence and history inside this same document.